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Recitation: 7

KNN Classifier

Action Scene	Romance Scene	Avg Duration	Happy Ending	Genre
15	2	3.5	0	A
1	12	4.2	1	R
18	1	3.8	0	A
2	14	4.5	1	R
14	3	3.2	1	A

A = Action, R = Romance

Test Dataset

AS	RS	AD	H.E	AG
10	5	3.7	1	A

Distance (Euclidean)

$$\sqrt{(x_{T1}-x_1)^2+(x_{T2}-x_2)^2+(x_{T3}-x_3)^2}$$

$d_1 = \sqrt{5^2+3^2+(-1)^2} = 5.92$

$d_2 = 11.40$

$d_3 = 9.0$

$d_4 = 12.04$

$d_5 = 4.47$

No weight updation

$\left[w = \frac{1}{d}\right]$

K = 5 means 5 elements

$w_1 = \frac{1}{d_1} = 0.192$

$w_2 = 0.08$, $w_3 = 0.11$ $w_4 = 0.08$

$w_5 = 0.22$

weights of A = $w_1 + w_3 + w_5$ = 0.52

weights of R = $w_2 + w_4$ = 0.08 + 0.08

0.16

∴ More is Action

KNN Regression

Age (x_1)	Previous Value (x_2)	Car Fee
2	3	2
3	4	3
5	1	0.5

Test : (4, 2, Y) — (2, 3), (3, 4), (5, 1)

$D_1 = \sqrt{(2-4)^2 + (3-2)^2} = 2.24$

$D_2 = \sqrt{(3-4)^2 + (4-2)^2} = 2.24$

$D_3 = \sqrt{(5-4)^2 + (1-2)^2} = \sqrt{2} = 1.41$

$\hat{y} = \frac{1}{K}\sum_{i=1}^{K} y_i \Rightarrow (3 + 0.5)/2 = 1.75$